

The Human-AI Partnership Framework

Where human judgment belongs

PCF 2.0 — Develop and Manage Products and Services

HITL Maturity Assessment — All Processes

L3 PROCESSES 12	L4 ACTIVITIES 65	M1 BASELINE H% 80%
M3 DESIGN INTENT H% 59%	M5 LEADING H% 36%	Δ M1 → M3 -21%

This document reports HITL (Human-in-the-Loop) maturity ratios for all of PCF 2.0 (Develop and Manage Products and Services) — 3 process groups, 12 Level 3 processes, and 65 Level 4 activities in total — at five maturity levels from M1 (Undesigned) through M5 (Leading), with M3 marking HITL Design Intent.

At M3 (HITL Design Intent), this domain's aggregate human-involvement ratio is **59%**, down from a **80%** undesigned (M1) baseline — a **-21-point** AI convergence opportunity. Mixed-pattern L3 rows reflect an equal-weighted rollup of their real L4 children rather than a force-classified single pattern.

HITL Maturity Assessment

H% = Human involvement at each maturity level · M3 = HITL Design Intent · Full L1→L2→L3→L4 hierarchy

PCF	L1-Domain / L2-Process Group / L3-Process / L4-Activity	Pattern	M1	M2	M3 ★	M4	M5	Δ M1→M3
2.0	Develop and Manage Products and Services · L1 DOMAIN AGGREGATE	—	80%	71%	59%	46%	36%	-21%
2.1	Govern and manage product/service development program · L2 AVG	—	78%	70%	58%	46%	36%	-20%
2.1.1	Manage product and service portfolio		87%	81%	70%	59%	50%	-17%
2.1.1.1	Evaluate performance of existing products/services against market opportunities	Knowledge	85%	78%	65%	52%	42%	-20%
2.1.1.2	Confirm alignment of product/service concepts with business strategy	Decision	92%	88%	80%	72%	65%	-12%
2.1.1.3	Prioritize and select new product/service concepts	Decision	92%	88%	80%	72%	65%	-12%
2.1.1.4	Plan and develop cost and quality targets	Knowledge	85%	78%	65%	52%	42%	-20%
2.1.1.5	Specify development timing targets	Knowledge	85%	78%	65%	52%	42%	-20%
2.1.1.6	Plan for product/service offering modifications	Knowledge	85%	78%	65%	52%	42%	-20%
2.1.2	Manage product and service life cycle		75%	66%	53%	40%	30%	-22%
2.1.2.1	Develop plan for new product/service development and introduction/launch	Knowledge	85%	78%	65%	52%	42%	-20%
2.1.2.2	Introduce new products/services	Transaction	60%	48%	35%	22%	12%	-25%
2.1.2.3	Retire outdated products/services	Transaction	60%	48%	35%	22%	12%	-25%
2.1.2.4	Identify and refine performance indicators	Knowledge	85%	78%	65%	52%	42%	-20%
2.1.2.5	Conduct post launch review	Knowledge	85%	78%	65%	52%	42%	-20%
2.1.3	Manage patents, copyrights, and regulatory requirements		87%	81%	68%	57%	47%	-19%
2.1.3.1	Conduct mandatory and elective reviews	Knowledge	85%	78%	65%	52%	42%	-20%
2.1.3.2	Review infringement of patents and copyrights	Exception	95%	92%	82%	75%	68%	-13%
2.1.3.3	Determine patent and copyright needs	Knowledge	85%	78%	65%	52%	42%	-20%
2.1.3.4	Define product technical	Knowledge	85%	78%	65%	52%	42%	-20%

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PCF	L1-Domain / L2-Process Group / L3-Process / L4-Activity	Pattern	M1	M2	M3 ★	M4	M5	Δ M1→M3
	documentation management requirements							
2.1.3.5	Manage regulatory requirements	Knowledge	85%	78%	65%	52%	42%	-20%
2.1.4	Manage product and service master data		64%	53%	40%	28%	18%	-24%
2.1.4.1	Manage materials master lists	Transaction	60%	48%	35%	22%	12%	-25%
2.1.4.2	Manage bills of material	Transaction	60%	48%	35%	22%	12%	-25%
2.1.4.3	Manage routings	Transaction	60%	48%	35%	22%	12%	-25%
2.1.4.4	Manage specifications	Transaction	60%	48%	35%	22%	12%	-25%
2.1.4.5	Manage drawings	Transaction	60%	48%	35%	22%	12%	-25%
2.1.4.6	Manage product/material classification	Transaction	60%	48%	35%	22%	12%	-25%
2.1.4.7	Develop and maintain quality/inspection documents	Document	72%	58%	40%	28%	18%	-32%
2.1.4.8	Maintain process specification data	Transaction	60%	48%	35%	22%	12%	-25%
2.1.4.9	Manage traceability data	Transaction	60%	48%	35%	22%	12%	-25%
2.1.4.10	Review and approve data access requests	Decision	92%	88%	80%	72%	65%	-12%
2.2	Generate and define new product/service ideas · L2 AVG	—	83%	76%	63%	50%	40%	-20%
2.2.1	Perform discovery research	Knowledge	85%	78%	65%	52%	42%	-20%
2.2.1.1	Identify new technologies	Knowledge	85%	78%	65%	52%	42%	-20%
2.2.1.2	Develop new technologies	Knowledge	85%	78%	65%	52%	42%	-20%
2.2.1.3	Assess feasibility of integrating new leading technologies into product/service concepts	Knowledge	85%	78%	65%	52%	42%	-20%
2.2.2	Generate new product/service concepts		80%	72%	59%	46%	36%	-21%
2.2.2.1	Gather new product/service ideas and requirements	Transaction	60%	48%	35%	22%	12%	-25%
2.2.2.2	Analyze new product/service ideas and requirements	Knowledge	85%	78%	65%	52%	42%	-20%
2.2.2.3	Evaluate new product/service inputs and requirements	Knowledge	85%	78%	65%	52%	42%	-20%
2.2.2.4	Formulate new product/service concepts	Knowledge	85%	78%	65%	52%	42%	-20%
2.2.2.5	Identify potential improvements to existing products and services	Knowledge	85%	78%	65%	52%	42%	-20%
2.2.3	Define product/service development requirements	Knowledge	85%	78%	65%	52%	42%	-20%
2.2.3.1	Define product/service requirements	Knowledge	85%	78%	65%	52%	42%	-20%
2.2.3.2	Define post launch support model	Knowledge	85%	78%	65%	52%	42%	-20%
2.2.3.3	Identify product/service bundling opportunities	Knowledge	85%	78%	65%	52%	42%	-20%
2.3	Develop products and services · L2 AVG	—	77%	68%	55%	42%	33%	-22%
2.3.1	Design and prototype products and services		80%	72%	59%	47%	37%	-21%
2.3.1.1	Assign resources to product/service project	Decision	92%	88%	80%	72%	65%	-12%
2.3.1.2	Prepare high-level business case and technical assessment	Knowledge	85%	78%	65%	52%	42%	-20%
2.3.1.3	Develop product/service design specifications	Knowledge	85%	78%	65%	52%	42%	-20%
2.3.1.4	Develop user experience design specifications	Knowledge	85%	78%	65%	52%	42%	-20%
2.3.1.5	Provide warranty-related recommendations	Knowledge	85%	78%	65%	52%	42%	-20%
2.3.1.6	Document design specifications	Document	72%	58%	40%	28%	18%	-32%
2.3.1.7	Conduct mandatory and elective external reviews	Knowledge	85%	78%	65%	52%	42%	-20%
2.3.1.8	Design products/services	Knowledge	85%	78%	65%	52%	42%	-20%
2.3.1.9	Build prototypes/proof of concepts	Transaction	60%	48%	35%	22%	12%	-25%
2.3.1.10	Develop and test prototype production and/or service delivery process	Transaction	60%	48%	35%	22%	12%	-25%
2.3.1.11	Eliminate quality and reliability	Exception	95%	92%	82%	75%	68%	-13%

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	problems							
2.3.1.12	Conduct in-house product/service testing and evaluate feasibility	Transaction	60%	48%	35%	22%	12%	-25%
2.3.1.13	Identify design/development performance indicators	Knowledge	85%	78%	65%	52%	42%	-20%
2.3.1.14	Collaborate on design with suppliers and external partners	Knowledge	85%	78%	65%	52%	42%	-20%
2.3.2	Test market for new or revised products and services		79%	70%	58%	44%	34%	-21%
2.3.2.1	Prepare detailed market study	Knowledge	85%	78%	65%	52%	42%	-20%
2.3.2.2	Conduct customer tests and interviews	Transaction	60%	48%	35%	22%	12%	-25%
2.3.2.3	Finalize product/service characteristics and business cases	Knowledge	85%	78%	65%	52%	42%	-20%
2.3.2.4	Finalize technical requirements	Knowledge	85%	78%	65%	52%	42%	-20%
2.3.3	Prepare for production/service delivery		65%	54%	41%	28%	18%	-24%
2.3.3.1	Design and obtain necessary capabilities/materials and equipment	Transaction	60%	48%	35%	22%	12%	-25%
2.3.3.2	Identify requirements for changes to manufacturing/delivery processes	Knowledge	85%	78%	65%	52%	42%	-20%
2.3.3.3	Request engineering/process change	Transaction	60%	48%	35%	22%	12%	-25%
2.3.3.4	Install and validate production/service delivery process	Transaction	60%	48%	35%	22%	12%	-25%
2.3.3.5	Validate launch procedures	Transaction	60%	48%	35%	22%	12%	-25%
2.3.4	Define warranty claims		76%	66%	52%	40%	31%	-24%
2.3.4.1	Determine and document warranty policies	Document	72%	58%	40%	28%	18%	-32%
2.3.4.2	Create and manage warranty rules/claim codes for products	Transaction	60%	48%	35%	22%	12%	-25%
2.3.4.3	Agree on warranty responsibilities with suppliers	Decision	92%	88%	80%	72%	65%	-12%
2.3.4.4	Define warranty related offerings for customers	Knowledge	85%	78%	65%	52%	42%	-20%
2.3.4.5	Communicate warranty policies and offerings	Document	72%	58%	40%	28%	18%	-32%
2.3.5	Develop recall strategy	Knowledge	85%	78%	65%	52%	42%	-20%

■ Decision ■ Knowledge ■ Document ■ Transaction ■ Exception

Full L1→L2→L3→L4 hierarchy · Mixed-pattern rows show an equal-weighted rollup of their real L4 children · © Timothy P. King & Claude (Anthropic) 2026